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Experiments and Causal Inference | Final Project Report  
June 21, 2026

# The Effects of Short-Term Rental Platforms on the Housing Market

*Evidence from Saint Lucia Using a Bartik Shift-Share Instrumental Variable Approach*

## 1. Introduction

Since the late 2000s, digital short-term rental (STR) platforms such as Airbnb, Vrbo, and Booking.com have enabled property owners to convert residential housing into tourist accommodation. When tourist rentals generate higher returns than long-term leasing, landlords have an incentive to withdraw units from the residential market, reducing housing supply and increasing affordability pressures. Studies from cities such as Barcelona, Boston, and New York City find that STR growth is associated with higher rents and lower housing availability, yet evidence remains concentrated in North America and Western Europe. Small Island Developing States (SIDS), where tourism dependence is often greater and data more limited, remain understudied.

Saint Lucia provides an important case. Tourism is a major economic sector, and STR platform usage expanded rapidly after 2014, particularly in tourism-intensive constituencies such as Gros Islet and Soufrière. Unlike many larger cities, Saint Lucia lacks rent stabilization, STR licensing enforcement, and extensive tenant protections, making the housing market potentially more vulnerable to tourism-driven demand shocks.

This report uses Saint Lucian census microdata and a causal identification strategy to examine four questions: (RQ1) Do renters in tourism-intensive areas face higher rents? (RQ2) Has STR expansion reduced renter-occupied housing availability in high-tourism constituencies? (RQ3) Is any shift in tenure associated with the replacement of renters by wealthier owner-occupiers? (RQ4) Is STR expansion associated with increased household crowding among remaining renters? The objective is to estimate the causal effect of STR platform expansion on housing tenure and affordability in a small island developing state, and in doing so to test whether the displacement patterns documented in large mainland cities also hold in a tourism-dependent island context.

## 2. Research Design

### 2.1 Identification Strategy

The central identification problem is endogenous platform placement. STR units did not appear randomly: they concentrated where tourism demand was already strong, so a regression of renter outcomes on observed STR counts would recover pre-existing locational advantage rather than a causal platform effect. I address this with a Bartik (shift-share) instrumental variable (Bartik, 1991; Goldsmith-Pinkham, Sorkin, and Swift, 2020) that isolates variation in tourism exposure driven by a fixed, pre-platform amenity endowment interacted with a national tourism trend.

The instrument multiplies a share component by a shift component, summarised in Table 1.

**Table 1. Components of the Bartik shift-share instrument.**

| Component                            | Description                                          | Measurement                                                                                                                        |
|--------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Share (NTEc)</b>                  | Constituency c's pre-existing natural attractiveness | Weighted score of beaches, peaks, waterfalls, volcanic features, and nature reserves from OpenStreetMap (fixed before the STR era) |
| <b>Shift (<math>\Delta g</math>)</b> | National growth in tourism after platform launch     | 18.7% increase in stay-over arrivals from the pre-STR average (2010–2014) to the post-launch average (2015–2019, 2022)             |

The instrument is the three-way interaction, switched on only in the post period:

$$Instrument = NTEc \times \Delta g \times Post_t$$

where  $Post_t$  is an indicator equal to 1 in 2022 and 0 in 2010. The shift-share logic holds that constituencies with more natural beauty (higher NTE) should absorb a larger share of the national tourism boom after 2014. Soufrière, with the iconic Pitons and the Sulphur Springs, has a high NTE; Dennery North, with no marquee natural features, has a low one. Because the natural features existed decades before Airbnb, they cannot have been caused by post-2014 housing-market changes, providing a plausible source of exogenous variation. The identifying assumption is that, conditional on fixed area differences and common time shocks, NTE affects housing outcomes only through this tourism channel and is uncorrelated with pre-existing trends in renter rates, an assumption probed directly in the robustness section.

Because the national arrivals shift ( $\Delta g$ ) is a single scalar applied identically to every constituency, it contributes no cross-sectional variation of its own; all identifying variation in the instrument comes from the NTE share. In this single-shift design the instrument is therefore equivalent to  $NTE \times Post$  scaled by a constant, and its credibility rests entirely on the exogeneity of the share rather than on the timing of the

shift. The arrivals growth serves to anchor the magnitude of the post-period tourism expansion and to motivate the share economically, not to drive identification.

The study estimates a three-equation system. The reduced form regresses the binary renter indicator on the instrument with constituency and year fixed effects:

$$\text{Reduced form: } Y_{ic_t} = \alpha + \beta(NTEc \times \Delta g \times Post_t) + \gamma_c + \delta_t + \varepsilon_{ic_t}$$

$$\text{First stage: } STRc_t = \pi(NTEc \times \Delta g \times Post_t) + \gamma_c + \delta_t + u_{ic_t}$$

$$\text{2SLS: } Y_{ic_t} = \alpha + \tau \cdot STRc_t + \gamma_c + \delta_t + \eta_{ic_t}$$

Here  $Y_{ic_t}$  is a binary indicator equal to 1 if household  $i$  in constituency  $c$  at time  $t$  rents its dwelling. The constituency fixed effects  $\gamma_c$  absorb all time-invariant differences across areas (for example, baseline urbanization), and the year fixed effect  $\delta_t$  absorbs any common island-wide shock between 2010 and 2022. Consequently,  $\beta$  identifies only the differential change in renting probability in high-NTE areas relative to low-NTE areas during the STR era.

## 2.2 Treatment and Outcome Variables

**Treatment variable.** The endogenous treatment is STR penetration, measured at the building level using the 2022 IDDetail building enumeration file. Two `dwell_status` codes identify short-term rental use a narrow measure of 245 flagged buildings, with a broad robustness measure described in Section 3.3. Because the building file is at the district level but the analysis is at the constituency level, STR units are disaggregated using each constituency's share of 2010 census household weights within its district.

**Outcome variables.** Four outcomes correspond to the research questions:

- Renter rate (RQ2 — Stock Depletion): binary indicator equal to 1 if the household rents (private or government), 0 otherwise.
- Log monthly rent (RQ1 — Rent Inflation): continuous measure of monthly rent in 2022 East Caribbean Dollars (EC\$), available only in the 2022 census.
- Owner-occupancy rate (RQ3 — Tenure Shift): binary indicator equal to 1 if the household owns outright or with a mortgage.
- Persons per bedroom (RQ4 — Crowding): continuous measure of household size divided by number of bedrooms.

## 2.3 Covariates and Controls

All regression specifications include constituency and year fixed effects (two-way fixed effects, TWFE). The rent model includes unit-level controls for number of bedrooms, household size, and persons per bedroom to absorb dwelling-quality differences; constituency fixed effects are excluded in this cross-section because NTE is a constituency-level variable and would be perfectly collinear with them. Census household weights are applied throughout. Standard errors are clustered at the geographic unit of analysis (constituency or district).

## 3. Data

### 3.1 Data Sources

The analysis combines five sources: two full-population census waves, a building-level enumeration file that flags STR units, the constituency NTE index, and national tourism statistics. Table 2 inventories them.

**Table 2. Data sources.**

| Source                           | Unit      | Key variables                                             | Years           |
|----------------------------------|-----------|-----------------------------------------------------------|-----------------|
| 2010 Population & Housing Census | Household | Tenure, bedrooms, household size, constituency, weight    | 2010 (pre-STR)  |
| 2022 Population & Housing Census | Household | Tenure, monthly rent, bedrooms, constituency, weight      | 2022 (post-STR) |
| IDDetail Building Enumeration    | Building  | dwell_status (STR flags), district code                   | 2022            |
| OpenStreetMap / Overpass API     | Geo-point | Beaches, peaks, waterfalls, volcanic features, viewpoints | Fixed, pre-2014 |
| CSO Tourism Statistics           | National  | Annual stay-over arrivals                                 | 2010–2022       |

Source: CSO Saint Lucia; OpenStreetMap / Overpass API (May 2026).

The 2010 and 2022 census microdata were obtained directly from the Central Statistical Office (CSO) and contain full-population household records.

### 3.2 Sample Construction and Tenure Harmonisation

The 2010 wave covers 50,241 households and the 2022 wave 50,588, each with constituency identifiers, tenure status, bedrooms, and household size; only 2022 collected monthly rent. Tenure categories differ across waves and are harmonised to three groups. Owner-occupied combines outright ownership and mortgage holders (2010 codes H13\_OWN = 1–2; 2022 codes h2\_3a = 1–6). Renter combines private and government tenants (2010 codes 3–4; 2022 codes 7–8). Rent-free households, typically family

arrangements, are 2010 code 5 and 2022 code 9. Both waves use 17 electoral constituencies and census household weights are applied throughout. The pooled household panel contains 100,829 observations. The rent analysis is restricted to the 5,643 renter-occupied households in 2022 with valid positive rent observations, while the crowding analysis uses 63,724 household observations with valid bedroom counts.

### 3.3 Measuring STR Units

The 2022 IDDetail file records the occupancy status of all 77,742 structures island-wide. Two dwell\_status codes identify platform activity: code 8 (“Short Term Occupation”, n = 162) and code 12 (“AirBNB”, n = 83), for 245 STR-flagged buildings in the narrow measure. A broad measure adds code 4 (“Vacant — Seasonally”, n = 1,148) for a robustness check; code 4 is excluded from the main measure because its distribution concentrates in low-NTE southern parishes, where seasonal vacancy more plausibly reflects labour migration and second-home patterns than platform activity. As an external benchmark, Airbtics reports 1,529 active listings island-wide in May 2026; the gap reflects four years of post-census growth and informal units below the enumerator threshold.

### 3.4 Natural Tourism Endowment (NTE)

The NTE index measures each constituency's concentration of natural geographic amenities relative to the rest of the island, built entirely from fixed physical features with no built infrastructure (no hotels, resorts, or guesthouses), so it cannot respond to the post-2014 accommodation market. Features are classified into six categories and weighted by their pull on leisure visitors: beaches 3.0, mountain peaks 2.5, volcanic features and hot springs 2.0, waterfalls and nature reserves 2.0, and scenic viewpoints 1.5. Each feature is assigned to a constituency by GPS coordinates; for districts with multiple constituencies, features are split by latitude relative to the district median, with verified overrides where local knowledge corrects the coordinate. Soufrière (24.7 percent) and Gros Islet (17.2 percent) dominate the endowment, while the five Castries constituencies and the Dennery parishes cluster at the bottom. The shares sum to 1.0 across all 17 constituencies.

Formally, the NTE share for constituency  $c$  is the constituency's total weighted feature score divided by the island-wide total, so that the shares sum to one:

$$NTE_c = \frac{\sum_{j \in c} w_j}{\sum_{c'} \sum_{j \in c'} w_j}$$

where  $j$  indexes the natural features assigned to constituency  $c$  and  $w_j$  is the feature-type weight defined above. One assignment case requires explicit treatment. Castries district contains five electoral

constituencies, but no public sub-district boundary shapefile exists to place individual features within it, so the district's combined weighted score is split equally across its five constituencies. Because all five Castries units sit near the bottom of the endowment distribution, this conservative equal split has little leverage on the estimates, and the leave-one-out and district-aggregation checks confirm the result does not hinge on it. The individual named features underlying every constituency score were cross-checked against Google Maps and local geographic knowledge rather than taken from raw OpenStreetMap tags alone, reducing the risk of misclassified or duplicated points.

**Table 3. Natural Tourism Endowment (NTE) by constituency.**

| <b>Constituency</b>   | <b>NTE share</b> | <b>Tourism tier</b> |
|-----------------------|------------------|---------------------|
| Soufrière             | 24.7%            | High                |
| Gros Islet            | 17.2%            | High                |
| Anse-la-Raye/Canaries | 9.8%             | Medium              |
| Laborie               | 9.6%             | Medium              |
| Babonneau             | 5.6%             | Medium              |
| Choiseul              | 4.7%             | Medium              |
| Vieux-Fort North      | 4.2%             | Medium              |
| Micoud North          | 3.7%             | Low                 |
| Castries North        | 3.3%             | Low                 |
| Micoud South          | 3.3%             | Low                 |
| Vieux-Fort South      | 3.0%             | Low                 |
| Castries South        | 2.3%             | Low                 |
| Dennerly South        | 2.1%             | Low                 |
| Dennerly North        | 2.1%             | Low                 |
| Castries South-East   | 2.1%             | Low                 |
| Castries East         | 1.2%             | Low                 |
| Castries Central      | 1.2%             | Low                 |

### 3.5 National Tourism Shift

The national shift collapses arrivals to a single growth factor: the pre-STR mean rises to the post-launch mean, a gain of 18.7 percent ( $\Delta g = 0.187$ ). Pandemic years 2020–2021 are excluded because COVID-19 cut arrivals by roughly 70 percent, which would otherwise understate the STR-era expansion.

**Table 4. National stay-over arrivals shift.**

| Period         | Years           | Mean arrivals                 |
|----------------|-----------------|-------------------------------|
| Pre-STR        | 2010–2014       | 316,385                       |
| Post-launch    | 2015–2019, 2022 | 375,610                       |
| Arrivals shift |                 | +18.7% ( $\Delta g = 0.187$ ) |

Three points justify entering the shift as a single growth factor rather than as a series of annual arrival counts. First, the housing outcomes are observed only in two census waves (2010 and 2022), so there are no intervening-year outcomes against which an annual arrivals series could be matched; the year-by-year variation in arrivals would have no corresponding dependent variable to identify. Second, in a two-period shift-share or difference-in-differences design, identification comes from the total change between baseline and follow-up, and the 18.7 percent figure represents the scale of the tourism shock to which the housing market adjusted. Third, averaging over the post-platform years (2015–2019 and 2022) rather than relying on a single year's count yields a more stable estimate of the new normal level of tourism, which is more plausibly the driver of structural housing change than a one-year spike; 2022 is retained because it is the census-match year and reflects recovered, structural tourism, while the pandemic years 2020–2021 are excluded for the reason given above.

### 3.6 Descriptive Patterns

The descriptive evidence aligns with the design before any regression is run. Ordering the 17 constituencies by NTE, the three highest-endowment areas, Soufrière, Gros Islet, and Anse-la-Rayé/Canaries, show the largest increases in renter rates between 2010 and 2022 (15.9, 15.6, and 7.2 percentage points, respectively), while low-NTE peripheral areas show smaller or negative changes. Owner-occupancy rates in these same constituencies fell by 40.7, 45.3, and 49.1 percentage points. Median monthly rents in 2022 are highest in Gros Islet (EC\$850) and lowest in Dennery North and Anse-la-Rayé/Canaries (EC\$300), though the rent gradient does not perfectly follow the NTE ranking due to urban employment concentration in Castries.

## 4. Methodology

### 4.1 Baseline Estimation Framework

The core estimating equations form a three-part system: a reduced form, a first stage, and a two-stage least squares (2SLS) combination, all estimated with two-way fixed effects on the pooled 2010–2022

household panel. The constituency effects  $\gamma_c$  absorb each area's baseline renter rate and the year effects  $\delta_t$  absorb any common shock, so  $\beta$  captures the differential change in renting probability between high- and low-NTE constituencies once the platform era begins. The first stage replaces the outcome with the STR unit count; the 2SLS coefficient is the ratio of the reduced-form to the first-stage coefficient. Standard errors are clustered at the geographic unit of analysis and census household weights are applied throughout.

Three specifications are estimated: 17 constituencies with the narrow STR measure (the primary specification), 17 constituencies with the broad measure (testing the code-4 classification decision), and 10 districts with the narrow measure (testing robustness to geographic aggregation). The rent question (RQ1) is treated differently. Because the 2010 census did not collect rent, there is no rent change to difference; the rent model is a 2022 cross-section of log monthly rent on NTE with dwelling controls, and it deliberately omits constituency fixed effects because NTE, being constituency-level, would be perfectly collinear with them. Table 5 summarises which design serves which question.

**Table 5. Estimation strategy by research question.**

| RQ  | Question                               | Method                            | Bartik used?           |
|-----|----------------------------------------|-----------------------------------|------------------------|
| RQ1 | Are rents higher in high-NTE areas?    | 2022 cross-section OLS            | No (no 2010 rent data) |
| RQ2 | Did renting fall in high-NTE areas?    | Reduced form + first stage + 2SLS | Yes — main result      |
| RQ3 | Did ownership rise in high-NTE areas?  | Reduced form (TWFE)               | Yes                    |
| RQ4 | Did crowding change in high-NTE areas? | Reduced form (TWFE)               | Yes                    |

*LPM = linear probability model. TWFE = two-way fixed effects. All Bartik specifications use the instrument  $NTE \times \Delta g \times Post$ .*

## 4.2 Weak-Instrument-Robust Inference

Because the first stage relies on a small number of geographic units (34 constituency-year cells), the first-stage F-statistics fall below the conventional threshold of 10, so standard 2SLS confidence intervals may be unreliable. I complement the 2SLS Wald intervals with the Anderson-Rubin (AR) test, which constructs a confidence set that is valid regardless of instrument strength by testing, for each hypothesized  $\tau$ , whether the instrument retains explanatory power after adjusting the outcome by  $\tau$ .



### 4.3 Robustness Checks

A battery of robustness checks addresses the small-N nature of the geographic design, outlined in Table 6.

**Table 6. Robustness checks.**

| Test                              | Purpose                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Broad STR measure</b>          | Tests sensitivity to including Code 4 (seasonal vacant) in the STR count                                                |
| <b>District-level aggregation</b> | Tests whether results are driven by constituency boundary definitions                                                   |
| <b>Randomization inference</b>    | Permutes NTE across constituencies 999 times to obtain an exact finite-sample p-value under the sharp null of no effect |
| <b>Leave-one-out</b>              | Re-estimates the reduced form dropping one constituency at a time to assess outlier influence                           |
| <b>Balance test</b>               | Regresses 2010 baseline housing characteristics on NTE to detect pre-treatment correlation with trends                  |
| <b>Anderson-Rubin CI</b>          | Provides weak-instrument-robust confidence intervals for the 2SLS estimates                                             |

The randomization inference test is particularly important in this setting because the 2001 census microdata required for a pre-period placebo regression (2001–2010), the most direct test of share exogeneity, are not currently accessible through the CSO public portal. The permutation test asks whether the observed spatial correlation between NTE and renter-rate changes could arise from a random assignment of natural endowments across constituencies. This is a data-availability constraint, not a methodological one: the estimation code is fully specified and the placebo can be activated if the 2001 file is obtained.

## 5. Results

### 5.1 RQ2: Stock Depletion — The Primary Causal Result

The causal claim of this report is narrow and specific: conditional on fixed area differences and common time shocks, greater pre-platform tourism-amenity exposure raised the probability of renting during the STR era. This claim rests on the exogeneity of NTE (the exclusion restriction), not on the strength of the first stage, and is therefore unaffected by the weak-instrument diagnostics reported below. The reduced form regresses the renter indicator directly on the instrument and never divides by the first stage, so a weak first stage cannot distort it; what the weak first stage does compromise is the scaled 2SLS magnitude, which is why that estimate is reported only for interpretability. The reduced form isolates the causal effect of tourism exposure during the STR era, not STR penetration directly. *Because the first-stage mapping to STR units is weak, the safer causal claim is that tourism endowments raised renter rates.*

**Reduced form.** Table 7 reports the reduced-form LPM results. The coefficient on the Bartik instrument is 2.106 (cluster SE = 0.586, Wald F = 12.9,  $p = 0.002$ ) in the 17-constituency specification. This indicates that a constituency at the island-wide mean NTE experienced a significantly larger increase in its renter rate during the STR era than a constituency with zero natural amenity endowment, after absorbing baseline levels and common year effects. In level terms, moving across the observed NTE range maps onto a clear, monotonic gradient in renter-rate growth. High-endowment constituencies (Soufrière, Gros Islet) saw renter rates rise ~16 percentage points from 2010–2022, while low-endowment areas saw none; the instrument pins this gap on differential tourism exposure, not island-wide trends. The district-level specification confirms robustness to geographic aggregation ( $\beta = 2.334$ , SE = 0.254,  $p < 0.01$ ), so the result is not an artefact of the 17-constituency geography.

**Table 7. Reduced-form LPM: Pr(renting) on NTE  $\times$  Arrivals Shift  $\times$  Post.**

|                                           | <b>(1) 17 Constituencies</b> | <b>(2) 10 Districts</b> |
|-------------------------------------------|------------------------------|-------------------------|
| NTE $\times$ Arrivals Shift $\times$ Post | 2.106***                     | 2.334***                |
| (cluster SE)                              | (0.586)                      | (0.254)                 |
| Constituency/District FE                  | Yes                          | Yes                     |
| Year FE                                   | Yes                          | Yes                     |
| N (household-years)                       | 100,829                      | 100,829                 |
| R <sup>2</sup>                            | 0.052                        | 0.043                   |

Notes: Outcome = 1 if household rents, 0 otherwise. SEs clustered at the geographic unit. \*\*\*  $p < 0.01$ . Census household weights.

**First stage and 2SLS.** Table 8 reports the first-stage and 2SLS results. The first-stage coefficients are large and positive, confirming the instrument's relevance directionally (NTE  $\times$   $\Delta g \times$  Post predicts where STR units concentrate), but the F-statistics fall below 10 in all specifications because the panel has only 34 cells (17 constituencies  $\times$  2 years). The 2SLS estimates are statistically significant but are reported for interpretability and read cautiously; the reduced form, which does not rely on the first stage, is the primary evidence, and Section 5.5 shows the weak first stage does not distort inference.

**Table 8. First stage and 2SLS, three specifications.**

|                                              | <b>(1) 17C Narrow</b> | <b>(2) 17C Broad</b> | <b>(3) 10D Narrow</b> |
|----------------------------------------------|-----------------------|----------------------|-----------------------|
| First stage: NTE $\times\Delta g\times$ Post | 1,154.783             | 3,896.697            | 1,194.032             |
| (SE)                                         | (890.386)             | (2,274.037)          | (726.131)             |
| First-stage F                                | 1.7                   | 2.9                  | 2.7                   |
| 2SLS: STR units (instr.)                     | 0.0011***             | 0.0004***            | 0.0018***             |
| (SE)                                         | (0.0002)              | (0.0001)             | (0.0002)              |
| AR 95% CI                                    | [0.0009, 0.0012]      | n/a                  | n/a                   |
| N (household-years)                          | 100,829               | 100,829              | 100,829               |

Notes: Narrow = Codes 8+12 ( $n = 245$ ); Broad = Codes 4+8+12 ( $n = 1,393$ ). AR = Anderson-Rubin confidence interval. \*\*\*  $p < 0.01$ .

**Randomization inference.** The permutation test assigns NTE values randomly across constituencies 999 times and re-estimates the reduced form for each draw. The exact two-sided p-value is 0.029, meaning fewer than 3 percent of random spatial assignments produce a coefficient as large as the observed 2.106. For a small-N design this is a more credible significance check than asymptotic clustered errors.

**Leave-one-out.** Re-estimating the reduced form on each of the 17 possible 16-constituency subsamples yields positive coefficients in all cases, with no single constituency driving the result. Dropping Gros Islet or Soufrière widens the confidence intervals but does not flip the sign.

## 5.2 RQ1: Rent Inflation

Because the 2010 census did not collect rent data, the rent outcome is estimated as a 2022 cross-section of log monthly rent on NTE with dwelling controls (Table 9). The NTE coefficient is stable across specifications but imprecisely estimated ( $p = 0.833$  in column 3). The point estimate implies an approximate 11 percent rent premium between the highest- and lowest-NTE constituencies, conditional on dwelling characteristics. While economically meaningful, this result is statistically underpowered with only 17 clusters and should be treated as suggestive; it is the report's weakest finding and would need a longer rent panel to confirm.

**Table 9. Log monthly rent on NTE, 2022 renter households.**

|                     | (1) No controls | (2) Unit controls | (3) + Crowding |
|---------------------|-----------------|-------------------|----------------|
| NTE                 | 0.563           | 0.454             | 0.449          |
| (SE)                | (2.059)         | (2.098)           | (2.092)        |
| Bedrooms            |                 | 0.316***          | 0.240***       |
| Household size      |                 | -0.038**          | 0.018          |
| Persons per bedroom |                 |                   | -0.111***      |
| Observations        | 5,643           | 5,643             | 5,643          |
| R <sup>2</sup>      | 0.003           | 0.085             | 0.087          |

Notes: SEs clustered by constituency ( $N = 17$  clusters). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ .

## 5.3 RQ3: Homeownership — Tenure-Shift Externalities

If STR pressure raised property values and attracted wealthier in-migrants, one might expect owner-occupancy to rise in high-NTE areas as wealthier buyers move in. The reduced-form results reject this hypothesis (Table 10). The owner-occupancy coefficient is negative:  $\beta = -1.975$  ( $p = 0.068$ ) at the constituency level and  $\beta = -2.726$  ( $p < 0.001$ ) at the district level. Renter rates rose while owner rates fell in the same high-NTE constituencies, which is inconsistent with a simultaneous investor-buyer-and-renter

sorting story. The more parsimonious reading is a single tenure-conversion channel: previously owner-occupied and informally held housing was converted into rental (and STR) stock as tourism pressure rose, rather than new owner-occupier demand being drawn in.

**Table 10. Reduced-form LPM: Pr(owner-occupied).**

|                             | (1) 17 Constituencies | (2) 10 Districts |
|-----------------------------|-----------------------|------------------|
| NTE × Arrivals Shift × Post | −1.975*               | −2.726***        |
| (cluster SE)                | (1.010)               | (0.401)          |
| N (household-years)         | 100,829               | 100,829          |

Notes: Outcome = 1 if owner-occupied. Geographic and year FE. \*  $p < 0.10$ , \*\*\*  $p < 0.01$ .

## 5.4 RQ4: Crowding — Secondary Market Distortions

The crowding result is not robust (Table 11). The constituency-level estimate on persons per bedroom is positive ( $\beta = 0.692$ ,  $p = 0.276$ ) while the district-level estimate is negative ( $\beta = -0.208$ ,  $p = 0.638$ ), so the sign flips with aggregation and neither reaches significance. This analysis does not provide reliable evidence of a crowding effect in either direction, so the displacement-via-crowding mechanism is unsupported.

**Table 11. Reduced-form OLS: persons per bedroom.**

|                             | (1) 17 Constituencies | (2) 10 Districts |
|-----------------------------|-----------------------|------------------|
| NTE × Arrivals Shift × Post | 0.692                 | −0.208           |
| (cluster SE)                | (0.614)               | (0.426)          |
| N (household-years)         | 63,724                | 63,724           |

Notes: Outcome = persons per bedroom. Geographic and year FE. Neither estimate is statistically significant.

## 5.5 Robustness Summary

Four checks support the main renter-rate result, all reported in Section 5.1: the broad STR measure and district-level aggregation both reproduce the positive, significant gradient; randomization inference returns an exact p-value of 0.029; leave-one-out returns positive coefficients in all 17 subsamples; and the Anderson-Rubin confidence set tracks the Wald interval, so the weak first stage does not distort inference.

A balance test of 2010 baseline housing characteristics on NTE returns no significant correlations, and even a significant level correlation would not threaten the design, because the constituency fixed effects absorb all time-invariant differences. What would threaten identification is a correlation between NTE and

pre-existing trends in renter rates. The ideal test, a 2001–2010 pre-period placebo regression, cannot currently be run because the 2001 census cross-tabulations are not exposed in the public CSO portal. In the interim the permutation test addresses the same identifying assumption by asking whether the observed spatial pattern of NTE is unusual relative to random assignment.

## 6. Discussion and Conclusion

### 6.1 Summary of Findings

The results point to a single dominant mechanism: rising renter rates in tourism-exposed constituencies rather than the gentrification pattern commonly observed in large cities. High-tourism constituencies experienced the largest increases in renter populations and the highest rents, while owner-occupancy declined and crowding showed no consistent change. The most credible causal evidence comes from the Bartik reduced-form estimate ( $\beta = 2.106$ ,  $p = 0.002$ ), which remains robust to randomization inference, leave-one-out tests, and alternative geographic aggregations. Although the first-stage results are weak and require caution when interpreting 2SLS estimates, the reduced-form findings remain informative.

The pattern differs from mainland displacement models but remains consistent with the broader literature linking STR expansion to higher rents. The estimated rent premium of approximately 11 percent is larger than many city-based estimates, consistent with the expectation that small island housing markets are less able to absorb demand shocks. Falling owner-occupancy alongside rising renting suggests a tenure-conversion process in which existing housing stock shifts toward rental use rather than being acquired by wealthier owner-occupiers. The absence of a crowding effect suggests that any displacement is not occurring through increased household densification.

The Saint Lucian case diverges from mainland studies due to distinct market structures. In Barcelona (García-López et al., 2020) and Boston (Horn and Merante, 2017), STR pressure operates in large, segmented urban markets where incoming capital displaces renters and attracts wealthier buyers, causing both rents and ownership to rise. Saint Lucia departs on two structural margins. First, its housing stock is small, owner-built, and informally held; rising tourism demand triggers the conversion of owner-occupied units into rentals and STRs rather than new owner acquisitions, driving renting up and ownership down (consistent with Wachsmuth and Weisler, 2018). Second, the absence of rent stabilization, STR licensing enforcement, and tenant protections removes institutional frictions, causing a faster, more complete tenure response. Rather than contradicting mainland literature, Saint Lucia reveals a different equilibrium produced by identical pressures in a small, lightly regulated, tourism-dependent market.

### 6.2 Limitations

Three limitations should be acknowledged. First, tourism-intensive constituencies are also major employment centres, making it difficult to fully separate STR effects from broader economic growth and labour-market attraction. Second, the absence of publicly available 2001 census microdata prevents a pre-period placebo test, limiting direct assessment of the exclusion restriction. Third, the STR measure is based on a 2022 snapshot and cannot capture changes in platform activity over time. Future research incorporating employment records and longitudinal platform data would strengthen causal identification.

### 6.3 Policy Implications

The findings suggest that policy should target housing conversion incentives rather than tourism activity itself. Priority measures include:

1. Allocating tourism levy revenues toward affordable housing in high-tourism constituencies.
2. Conditioning tourism-related tax incentives on maintaining primary residence in Saint Lucia.
3. Providing tax relief for long-term rental income to reduce incentives for STR conversion.
4. Encouraging visitor dispersion toward less tourism-intensive constituencies.
5. Implementing mandatory STR registration and stronger enforcement.
6. Establishing a landlord-tenant tribunal and improving access to platform listing data.

### 6.4 Conclusion

This study provides a rare causal analysis of STR platform effects on housing markets in a Caribbean Small Island Developing State. Using a Bartik shift-share IV design and census microdata, it finds that tourism-endowed constituencies experienced significant increases in renter rates, declining owner-occupancy, and no healthy changes in crowding during the STR era. These findings suggest that platform-driven tourism intensified pressure on local housing markets primarily through tenure conversion rather than classic gentrification. The results offer important evidence for housing policy in Saint Lucia and other tourism-dependent island economies, while highlighting the need for future research using longitudinal platform and employment data.

## References

- Barron, K., Kung, E., & Proserpio, D. (2021). The effect of home-sharing on house prices and rents: Evidence from Airbnb. *Marketing Science*, 40(1), 23–47.
- Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* Kalamazoo, MI: W.E. Upjohn Institute for Employment Research.
- Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1), 181–213.
- García-López, M.-À., Jofre-Monseny, J., Martínez-Mazza, R., & Segú, M. (2020). Do short-term rental platforms affect housing markets? Evidence from Airbnb in Barcelona. *Journal of Urban Economics*, 119, 103278.
- Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.
- Gotham, K. F. (2005). Tourism gentrification: The case of New Orleans' Vieux Carré (French Quarter). *Urban Studies*, 42(7), 1099–1121.
- Horn, K., & Merante, M. (2017). Is home sharing driving up rents? Evidence from Airbnb in Boston. *Journal of Housing Economics*, 38, 14–24.
- Pelling, M., & Uitto, J. I. (2001). Small island developing states: Natural disaster vulnerability and global change. *Environmental Hazards*, 3(2), 49–62.
- Saint Lucia Central Statistical Office. (2022). *2022 Population and Housing Census: Preliminary Report*. Castries: CSO.
- Wachsmuth, D., & Weisler, A. (2018). Airbnb and the rent gap: Gentrification through the sharing economy. *Environment and Planning A: Economy and Space*, 50(6), 1147–1170.
- Zervas, G., Proserpio, D., & Byers, J. W. (2017). The rise of the sharing economy: Estimating the impact of Airbnb on the hotel industry. *Journal of Marketing Research*, 54(5), 687–705.